Applied Business Statistics – Spring 2026
Chapter 4: Exercises

How the workshops work
You solve the exercises in your study group. Begin by discussing the answer to each exercise in the group. Group discussion increases your understanding of the concepts and helps you remember them better. It is important that you discuss examples so that you gain an understanding of the practical application of the theoretical concepts.

After you have discussed the exercise in the group, each of you should individually write down your answer in your own words. This is good practice for the exam, which is written and individual. The workshop instructors are present to help you if you have problems with an exercise. Any exercises that you do not manage to solve during the workshop become homework for next time.

4.1
Last year, the number of sick days due to colds and flu was recorded in a sample of 15 adults. The data are:

5, 7, 0, 3, 15, 6, 5, 9, 3, 8, 10, 5, 2, 0, 12

Calculate the mean, median, and mode. Explain the results.

4.3
The professors at Wilfrid Laurier University must submit their final exams to the secretary at least 10 days before the end of the semester. The exam coordinator took a sample of 20 professors and recorded the number of days before the final exam when the professors submitted their exams. The results are:

14, 8, 3, 2, 6, 4, 9, 13, 10, 12, 7, 4, 9, 13, 15, 8, 11, 12, 4, 0

Calculate the mean, median, and mode. Explain the results.

4.6
Return on Investment, ROI, over a 5-year period was as follows:

Year	1	2	3	4	5
ROI	0.10	0.22	0.06	-0.05	0.20

a) Calculate the mean and median of the returns. Explain the result.
b) Calculate the geometric mean.
c) Which of the three statistics calculated in parts a) and b) best describes the return over the 5-year period? Explain your answer.

4.12
The amount of time residents of Paris spend commuting was recorded in a sample of 235 commuters. Calculate the mean and median using the dataset **Xr04-12** in Python. See Example 4.10. Explain the results.

4.26
Calculate the variance and standard deviation in the following sample. Explain the result.

12, 6, 22, 21, 23, 13, 15, 17, 21

4.37
Many traffic experts argue that the most important factor in accidents is not the average speed, but the amount of variation in speed. On a highway section where many accidents occur, speed was measured in a sample of 200 vehicles. Using the dataset **Xr04-37**, calculate the variance and standard deviation of the speed in Python. See Example 4.36. Interpret the results.

4.54
Determine the first quartile, second quartile, and third quartile in the following data, and draw a box plot. Explain the results.

2, 4, 6, 8, 10, 12, 14, 16, 18, 20

4.71
The amount of time that guests spend over a cup of coffee or dessert negatively affects many restaurants' profits. To learn more about this variable, a sample of 200 groups of restaurant guests was observed, and the amount of time they spent in the restaurant was recorded.

Use the dataset **Xr04-71** in Python. See Example 4.67.

a) Determine the quartiles in the dataset.
b) What can you conclude about the amount of time spent in the restaurant?

4.81
Refer to Exercise 4.80 in the book. You now know that the two standard deviations are 16 and 12.

a) Calculate the correlation coefficient, **r**. What can you conclude from this statistic about the relationship between the two variables?
b) Calculate the coefficient of determination, $\mathbf{r^2}$, and describe what it shows about the relationship between the two variables.

4.83
Is the grade in a subject related to the amount of time spent studying? To investigate this possibility, a student took a random sample of 10 students who had Accounting last semester. The student asked

each of them to report their grade in the subject as well as the total number of hours they spent studying for the subject.

The data can be found in the dataset **Xr04-83**.

Calculate the following in Python. See Example 4.82.

a) Calculate the covariance.
b) Calculate the correlation coefficient, **r**.
c) Calculate the coefficient of determination, $\mathbf{r^2}$.
d) Use the least squares method to create a trend line.
e) What do the calculated statistics tell you about the relationship between grade and study time?

4.130
Refer to Exercise 3.54. See the Python example from Workshop 2. Use the dataset **Xr03-54** to calculate the following in Python:

a) Calculate the coefficients of the trend line. Find **a** and **b** in the equation:

y = ax + b

b) Calculate the coefficient of determination, $\mathbf{r^2}$.

Additional Exercise

4.68
The university's career counseling center wanted to know more about starting salaries among the university's graduates. They asked each graduate to report the highest salary offer they had received.

In the questionnaire, graduates also had to report their degree:

- Column A = BA, Bachelor of Arts
- Column B = BSc, Bachelor of Science
- Column C = BBA, Bachelor of Business Administration
- Column D = Other

Using the dataset **Xr04-68**, calculate the quartiles for each degree and explain what they show about the four groups of graduates.